INCH-POUND

MIL-C-44328A
11 July 1990
SUPERSEDING
MIL-C-44328
29 March 1988

MILITARY SPECIFICATION

CHICKEN CACCIATORE, THERMOSTABILIZED, TRAY PACK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers chicken cacciatore thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:28Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-44340 - Can, Tray Pack

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Poultry Products Inspection Regulations (9 CFR Part 381)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

U.S. Standards for Grades of Canned Mushrooms
U.S. Standards for Grades of Tomato Paste

(Application for copies should be addressed to the Director, Fruit and Vegetable Division, Agricultural Marketing Service (AMS), U.S. Department of Agriculture, Room 2077, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:28Z
Check the source to verify that this is the current version before use.

United States Standards for Condition of Food Containers

(Copies are available from the Chairman, Condition of Container Committee, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2506, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS)

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, 650 California Street, Suite 800, San Francisco, CA 94108.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3330 - Peel Adhesion of Pressure-sensitive Tape

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:28Z
Check the source to verify that this is the current version before use.

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 2200 Wilson Boulevard, Suite 400-CD, Arlington, VA 22201-3301.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Chicken. The chicken shall be prepared from chilled or frozen ready-to-cook fowl (mature female chicken) which has been processed in accordance with USDA Poultry Products Inspection Regulations. All chicken (raw or cooked) shipped between plants shall be accompanied by a USDA Poultry Products Grading Certificate to certify class and condition of the product and either the initial chilling date or initial freezing (in-storage) date.

3.2.1.1 Chicken, chilled. Raw bone-in or boneless chicken received in the chilled state shall not have been previously frozen and shall have been held at a temperature not to exceed 40°F for a period of time not to exceed 4 days following initial chilling and prior to preparation and further processing (see 3.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:28Z
Check the source to verify that this is the current version before use.

3.2.1.2 Chicken, frozen. Raw bone-in or boneless chicken received in the frozen state shall have been held at a temperature not to exceed 0°F for a period of time not to exceed 120 days following initial freezing and prior to preparation and further processing (see 3.3).

3.2.1.3 Chicken, cooked, frozen. Frozen cooked chicken shall be prepared from chilled chicken that complies with 3.2.1 and 3.2.1.1. Chicken shall be processed in accordance with 3.3.1, 3.3.1.1 or 3.3.1.2. Cooked chicken, in log, piece, or dice form, shall be packaged and vacuum-sealed in water-impermeable material having an oxygen permeability rate of not more than 10cc of oxygen per square meter per 24 hours at 73°F and 0 percent relative humidity and shall be frozen to an internal temperature of 0°F or below within 72 hours. The packaged, frozen cooked chicken shall have been held at 0°F or below for a period not to exceed 60 days prior to can filling (see 3.4). Frozen cooked chicken compliance with processing and packaging requirements shall be certified by the ingredient supplier. Alternatively, chicken dices produced from cooked chicken logs may be packaged in suitable material other than that described above and placed in a freezer immediately after dicing and frozen to an internal temperature of 0°F or below within 24 hours after placement in the freezer. Cooked, individually quick frozen chicken dices may also be packaged in other than the impermeable material and placed in a freezer immediately after packaging. Cooked, frozen chicken dices packaged in other than impermeable material shall be held at an internal temperature of 0°F for a period of time not to exceed 14 days. If other than impermeable packaging material is used, the type of material shall be approved by USDA.

3.2.2 Broth, chicken. Chicken broth may be frozen or canned (thermostabilized). The broth shall be inspected in accordance with USDA Poultry Products Inspection Regulations. It shall be free from extraneous material and cracklings. If frozen, broth shall have been held at 0°F or below for a period not to exceed 75 days prior to use in product formulation. The broth shall be clear, essentially fat free, and concentrated to a soluble solids level sufficient to comply with the solids requirement in the product formulation. The broth shall have a characteristic mild chicken broth odor and flavor and may contain flavor enhancers approved by FDA.

3.2.3 Tomatoes, canned, crushed. Canned tomatoes shall be peeled, cored, mature crushed tomatoes. The use of safe and suitable firming and acidification ingredients and salt is permitted. The canned tomatoes shall have not less than 8.0 percent tomato soluble solids and shall possess a red flesh color, normal character, and a good distinct acid sweet tomato flavor and odor. The crushed tomatoes shall be free of extraneous vegetable material and objectionable core material and skins (peel). The crushed tomatoes shall be from the latest date of pack.

3.2.4 Mushrooms, canned, pieces and stems. The canned pieces and stems shall be U.S. Grade B or better, in accordance with the U.S. Standards for Grades of Canned Mushrooms. The mushrooms shall be drained prior to use in the sauce formulation. The brine shall be retained for inclusion into the sauce. The canned mushrooms shall be from the latest date of pack.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:28Z
Check the source to verify that this is the current version before use.

3.2.5 Mushrooms, frozen, sliced. Frozen mushrooms shall be prepared from fresh, sound, properly trimmed and washed mushrooms possessing the characteristics of the species Agaricus (Psalliota) bisporus or Agaricus bitorguis. The mushrooms shall be of the white or cream color type and shall be of a randomly sliced style with approximately 1/8 inch slices. The frozen mushrooms shall be of the latest season's pack.

3.2.6 Tomato paste. Tomato paste shall be U.S. Grade A of the U.S. Standards for Grades of Tomato Paste.

3.2.7 Fat, chicken, rendered, frozen. Frozen rendered chicken fat shall have a clear, yellow color and a characteristic mild chicken-fat odor and flavor. The fat may contain USDA approved antioxidants. The chicken fat shall have a peroxide value not to exceed 6 meg./kg, a free fatty acid value not to exceed 0.50 percent (expressed as oleic acid), and a moisture content not to exceed 0.25 percent. The chicken fat shall be produced in a USDA inspected plant in accordance with USDA regulations and shall have been held at 0°F or below for a period not to exceed 75 days prior to formulation.

3.2.8 Starch, waxy maize, modified. Starch shall be white, odorless, finely pulverized, modified waxy maize starch for use in thermostabilized foods. The modified starch shall demonstrate initial viscosity development in the temperature range of 140° to 170°F and a typical viscosity (be fully hydrated) at common retort temperatures. The starch shall resist breakdown at low pH under shear stress and under conditions of cold storage. The starch shall be bland with essentially no cereal or starch taste and shall comply with MIL-STD-900.

3.2.9 Onions, dehydrated, chopped. Dehydrated chopped onions shall be Fancy Grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.10 Salt. Salt shall be noniodized, white, refined sodium chloride with or without anticaking agents and shall comply with purity standards for sodium chloride of the Food Chemicals Codex.

3.2.11 Cheese, parmesan, grated. The parmesan cheese shall conform with FDA standards of identity for parmesan cheese (21 CFR 133. 165). The cheese shall be grated parmesan cheese which has been aged 10 months or more. The grated cheese shall be free-flowing and free from lumps that do not fall apart under light finger pressure.

3.2.12 Sugar. Sugar shall be refined, sucrose, liquid or granulated, cane or beet sugar or a combination thereof, and shall comply with MIL-STD-900.

3.2.13 Peppers, green, sweet, freeze-dried. The peppers shall be 1/4 inch dices of sweet bell peppers which have been dehydrated by freeze-drying or partially air-dried followed by freeze-drying. The dices shall be practically free of seeds, seed cells, and diseased or rotted areas. The pepper dices shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:28Z
Check the source to verify that this is the current version before use.

have a fresh, clean, typical sweet green pepper flavor and odor, and the color shall be green with minimal brownish shade. The moisture content shall be not greater than 4 percent.

3.2.14 Lecithin. Lecithin shall comply with the Food Chemicals Codex.

3.2.15 Pepper, white, ground. Ground white pepper shall be derived from the dried mature berries of Piper nigrum L. from which the outer covering or the outer and inner coverings have been removed. The pepper shall have a characteristic penetrating odor, a hot biting pungent flavor, and a light color. The white pepper shall contain not less than 1.0 mL of volatile oil per 100 grams of ground white pepper and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 40 sieve.

3.2.16 Garlic powder. Garlic powder shall be Fancy Grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.17 Oregano, ground. Ground oregano shall be derived from the dried leaves of Origanum vulgare L. and shall possess a strong camphoraceous aroma and a pungent, slightly bitter flavor. The ground oregano shall contain not less than 2.0 mL of volatile oil per 100 grams of ground oregano and shall be of such size that 95 percent shall pass through a U.S. Standard No. 30 sieve.

3.2.18 Bay leaves, ground. Ground bay leaves shall be derived from the dried leaves of Laurus nobilis L. The bay leaves shall possess a pleasant aromatic odor, a pungent, mildly bitter flavor, and a pale green to yellow green color. A minimum of 95 percent shall pass through a U.S. Standard No. 30 sieve. Volatile oil content shall be not less than 1.0 mL of volatile oil per 100 grams of ground bay leaves.

3.2.19 Celery seed, ground. Ground celery seed shall be derived from the seed of Apium graveolens L. and shall be light to rich-brown in color, and shall possess a characteristic celery odor and flavor with a warm, slightly bitter aftertaste. The ground celery seed shall contain not less than 2.0 mL of volatile oil per 100 grams and shall be of such size that not less than 95 percent, by weight, shall pass through a U.S. Standard No. 35 sieve.

3.2.20 Fennel ground. Ground fennel shall be derived from the ripe fruit of Foeniculum vulgare Mill. The ground fennel shall be light green-brown to yellow-brown in color and shall possess a pleasant aromatic odor with a flavor resembling that of anise. The fennel shall contain not less than 1.0 mL of volatile oil per 100 grams of ground fennel and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 25 sieve.

3.2.21 Sodium tripolyphosphate. Sodium tripolyphosphate shall comply with the Food Chemicals Codex.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:28Z
Check the source to verify that this is the current version before use.

3.2.22 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the individual requirements specified in this document. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formula.

3.2.23 Water. Water used for ice-making, formulation, rehydration, and washing shall conform to the National Primary Drinking Water Regulations.

3.3 Preparation and further processing. Processing shall be on a continuous basis.

3.3.1 Chicken preparation. The chicken shall be made boneless and skinless. The wing tips, giblets, neck, feet, and viscera shall be excluded. The boned meat shall be free of pin feathers, skin, blood clots, bruises, blemishes, ligaments, tendons, coarse connective tissue, cartilage, and bone greater than 0.3 inch in any dimension. The boneless meat shall be composed of at least 50 percent white meat. Cooked chicken dices shall be prepared in accordance with 3.3.1.1 and 3.3.1.2.

3.3.1.1 Preparation of individually quick frozen (IQF) chicken dices. The IQF chicken dices shall be prepared as follows:

a. The chicken shall be cooked either before or after deboning by any commercially acceptable cook method in accordance with USDA Poultry Products Inspection Regulations.

b. The cooked chicken meat shall be cooled to an internal temperature of 28° to 40°F within 4 hours following the cooking process and shall be held not more than 48 hours at this temperature.

c. The cooked chicken meat shall be mechanically diced with dicer settings at 3/4 by 3/4 by 3/4 inch dimensions. The diced meat shall be held not more than 4 hours at an internal temperature of 28° to 40°F from time of dicing to the beginning of the IQF process.

d. Immediately after the IQF process, the frozen chicken dices shall be screened to eliminate fines. The frozen and screened chicken dices shall be packaged and vacuum sealed in water-impermeable material having an oxygen permeability rate of not more than 10 cc of oxygen per square meter per 24 hours at 73°F and 0 percent relative humidity. Frozen, cooked chicken meat shall not have been prepared from raw (uncooked) frozen chickens (see 3.2.1.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:28Z
Check the source to verify that this is the current version before use.

3.3.1.2 Chicken log preparation and processing. The chicken logs shall be prepared and processed as follows:

Ingredient	Percent by weight
Chicken meat	95.75
Ice or ice water	3.00
Salt	1.00
Sodium tripolyphosphate	0.25

a. Fresh chilled chicken meat processed on the day of boning shall be coarse ground within 24 hours following boning. If fresh-chilled (not previously frozen) chicken meat is transported between plants, it shall be maintained at an internal temperature of 28° to 40°F and coarse ground within 4 days following boning. Frozen chicken meat shall be maintained at an internal temperature of 0°F or lower until further processing and be coarse ground within 120 days after boning. The meat shall be mechanically coarse ground through a grinder plate having kidney-shaped openings measuring not less than 2 inches in the shortest dimension. The grinder shall be equipped with a 2-blade knife.

b. The coarse ground (chunked) chicken meat shall be mechanically mixed with the salt and sodium tripolyphosphate. The mixing shall continue until the mixture exhibits a sticky (tacky) consistency. Time from grinding to mixing shall not exceed 4 hours. The ground chicken meat shall be maintained at an internal temperature of 28° to 40°F during this time period.

c. The mixed meat shall immediately be mechanically stuffed tightly into prestuck, regenerated cellulose casings of a size to accommodate the finished product meat dimension requirement. The stuffed meat logs shall be in the cooking process within 4 hours after being stuffed. (This period of time may be extended up to 24 hours, provided the meat logs are maintained at an internal temperature of 28° to 40°F.)

d. The chicken logs shall be cooked in a cook house (smoke house without smoke) or by other commercially acceptable cook methods to provide a product meeting the finished product drained weight and other requirements.

e. Immediately after completion of the cooking process, the cooked meat logs shall be placed in an ice water bath, or cooled by other commercially acceptable cooling method, and rapidly cooled to an internal temperature of 130°F or below. The chicken logs shall be further cooled from 130° to 80°F or below within 2 hours and further cooled from 80° to 40°F or below within 4 hours. Cooked meat logs chilled to an internal temperature of 28° to 40°F shall be held for not more than 48 hours prior to dicing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:28Z
Check the source to verify that this is the current version before use.

f. The cooked chicken logs shall be mechanically diced with dicer settings at 3/4 by 3/4 by 3/4 inch dimensions. After dicing, the meat shall be screened to eliminate fines. The diced meat shall be held not longer than 24 hours at an internal temperature of 28° to 40°F until preparation of the finished product or, if stored in frozen form, until packaged and vacuum-sealed (see 3.2.1.3).

3.3.2 Sauce formulation and preparation. The sauce shall be formulated as follows:

Ingredients	Percent by weight
Chicken broth 1/	56.61
Tomato, crushed (8.0 percent tomato soluble solids) 2/	16.00
Mushrooms, stems, and pieces, canned, drained or mushrooms frozen	8.46
Tomato paste (26 percent natural tomato soluble solids) 2/	5.00
Chicken fat	4.00
Starch, waxy maize, modified	3.50
Onions, chopped, dehydrated 4/	3.45
Salt 3/	1.25
Cheese parmesan, grated	0.50
Sugar, granulated	0.50
Green pepper, diced, dehydrated 4/	0.30
Lecithin	0.20
Pepper, white, ground	0.09
Garlic powder	0.06
Oregano, ground	0.03
Bay leaf, ground	0.02
Celery seed, ground	0.02
Fennel seed, ground	0.01

1/ Chicken broth with different soluble solids content shall be adjusted to the 3.0 to 3.5 percent solids.

2/ If the specified tomato solids are not available, the tomato solids that are used must be adjusted to bring the solid level to that specified.

3/ The total amount of salt in sauce formula shall be adjusted as necessary to produce a product that complies with the finished product salt requirements.

4/ The dehydrated onions and green peppers shall be fully rehydrated in an excess of water, then thoroughly drained before incorporation into the sauce (see 3.3.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:28Z
Check the source to verify that this is the current version before use.

NOTE: The following sauce preparation procedures were used in the development of this product. Alternative procedures may be used provided finished product requirements are met. When alternative procedures are used, the time and temperature requirements specified for the prepared sauce are still applicable.

a. A slurry shall be made using part of the chicken broth, and the starch.

b. An emulsion shall be made by mixing the remainder of the chicken broth, mushrooms, crushed tomatoes, tomato paste, chicken fat, and lecithin. The emulsion shall be heated to a boil in a steam jacketed kettle with continuous and vigorous agitation to attain maximum emulsification of the fat.

c. To the emulsion, add the parmesan cheese and stir until the cheese is uniformly dispersed.

d. The emulsion shall be cooled to 180°F or lower and the remaining ingredients added. The emulsion shall be stirred to assure that the salt is dissolved, the dehydrated vegetables are rehydrated, and the spices are uniformly dispersed.

e. The starch slurry shall be added, and the final mixture heated to 180° to 190°F and held in this temperature range for 5 minutes.

f. The volume of the final mixture shall be adjusted with water to compensate for evaporation loss during heating and holding.

g. If the sauce is not to be immediately filled into cans, it shall be maintained in the temperature range of 150° to 180°F. The holding time from end of preparation until filling into the cans shall not exceed 4 hours.

3.3.3 Product preparation. The diced chicken and sauce shall be combined in the following proportions, using either a single or multi-stage filling operation:

Ingredient	Percent by weight
Chicken, cooked, diced	49.10
Sauce	50.90

3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with product such as to conform to the finished product requirements and to the following requirements:

a. Immediately after filling, each can shall be hermetically sealed under a vacuum established by a processing authority and specified in the scheduled process so as to ensure compliance with the finished product requirement (see 3.6t).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:28Z
Check the source to verify that this is the current version before use.

b. The filled and sealed tray pack cans shall be in the retort process within 2 hours after sealing.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting at 240°F or below until a sterilization value (F_o) of not less than 6.0 has been achieved.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odors or flavors such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

d. No individual can shall contain a bone piece measuring more than 0.3 inch in any dimension.

e. The total weight of skin, cartilage, coarse connective tissue, section of tendons or ligaments, and discolored meat, collectively, in a can shall not be more than 1 ounce.

f. There shall be no feathers or feather parts.

g. The average net weight shall be not less than 106 ounces.

h. No individual can shall contain less than 104 ounces of product.

i. The average drained weight of whole intact chicken pieces, mushrooms and other ingredients shall be not less than 59.0 ounces.

j. No individual can shall contain less than 54.0 ounces of drained, whole intact chicken pieces, mushrooms and other ingredients.

k. Of the 54.0 ounces drained weight of product, not less than 20.0 ounces shall be whole intact chicken pieces.

l. Texture of chicken pieces shall not be dry, rubbery, or mushy.

m. Texture of mushrooms shall not be mushy, hard, fibrous, or tough.

n. Sauce consistency shall be smooth, without lumps, and shall not be excessively thick or thin.

o. The average fat content shall be not greater than 7.0 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:28Z
Check the source to verify that this is the current version before use.

p. The fat content of an individual tray pack shall be not greater than 9.0 percent.

q. The salt content of an individual tray pack shall be not greater than 1.3 percent nor less than 0.5 percent.

r. The chicken pieces and sauce shall be uniformly distributed.

s. The product shall show no evidence of excessive heating (materially darkened or scorched).

t. Filled, sealed and retorted cans shall show evidence of proper vacuum as determined by concavity of the can lid (see 4.5.6).

3.6.1 Palatability. The product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The poultry component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under Poultry Products Inspection Regulations of the U.S. Department of Agriculture.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all the requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection and supervision necessary to ensure compliance with the requirements of this specification.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:28Z
Check the source to verify that this is the current version before use.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.5.3.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average net weight is less than 106 ounces or if the net weight of any individual can is less than 104 ounces, the lot shall be rejected.

4.5.3.2 Product inspection. The sample size shall be as indicated by the double sampling plan specified in table I. The sample cans shall be selected at random from the lot. The sample cans shall be heated in accordance with heating instructions on the can label, opened, and inspected for the defects listed in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:28Z
Check the source to verify that this is the current version before use.

TABLE I. Double sampling plan for product inspection 1/

Lot size (cans)	Sample size (cans)	Cumulative sample	Acceptance number	Rejection number
0 to 3,200	8	-	0	2
	8	16	1	2
3,201 to 35,000	13	-	0	3
	13	26	3	4

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE II. Product defects 1/ 2/ 3/

Category	Defect
Major	
101	Presence of bone piece measuring more than 0.3 inch in any dimension
102	Presence of feathers or feather parts
103	Total weight of skin, cartilage, coarse connective tissue, section of tendons or ligaments, and discolored meat, collectively, is more than 1 ounce
104	Drained weight of whole intact chicken pieces, mushrooms, and other ingredients in a can is less than 54.0 ounces 4/ 5/

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:28Z
Check the source to verify that this is the current version before use.

TABLE II. Product defects 1/ 2/ 3/ (cont'd)

Category	Defect
Major	
105	Of the 54.0 ounces drained weight of product, less than 20.0 ounces are whole intact chicken pieces 6/
106	Texture of chicken pieces dry, rubbery, or mushy 7/
107	Sauce consistency lumpy or is excessively thick or thin
108	Texture of mushrooms mushy, hard, fibrous, or tough 7/
109	Chicken pieces and sauce not uniformly distributed
110	Product shows evidence of excess heating (materially darkened or scorched)

1/ The presence of any foreign material (for example, dirt, insect parts, hair, wood, glass, metal), foreign odor or flavor (for example, burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an Agricultural Marketing Service (AMS) agent.)

3/ Interior of filled and sealed cans shall be examined visually for coating defects during product examination. Suspected rust spots shall be verified by testing in accordance with 4.5.7. Any spot verified as a rust spot shall be cause for rejection of the lot. Any scratch or fracture that penetrates through the coating of a can shall be cause for rejection of the lot.

4/ To determine drained weight, the free liquid in the can shall be poured off and the remaining contents shall be poured into a flat-bottom container. A minimum of three times the tray pack can's volume of 190° to 212°F water shall be added to the container so as to cover the contents. The contents and water shall be gently agitated such as to liquify rendered fat and to remove the sauce without breaking the chicken pieces, mushrooms or vegetables. The contents shall then be poured into a U.S. Standard 1/4-inch sieve in a manner that will distribute the product over the sieve without breaking the chicken pieces, mushrooms or vegetables. Sieve area shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:28Z
Check the source to verify that this is the current version before use.

such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at approximately a 45-degree angle and allowed to drain for 2 minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. The drained weight shall be reported to the nearest 0.1 ounce.

5/ The lot shall be rejected if the sample average drained weight of the whole intact chicken pieces, mushrooms, and other ingredients is less than 59.0 ounces.

6/ For purpose of this examination, the definition of a whole intact chicken piece is one that measures not less than 3/8 inch in each of two separate dimensions.

7/ Examination for appearance and texture conformance of the chicken and mushrooms shall be made of representative pieces immediately following the determination of drained weight.

4.5.3.3 Fat and salt content testing. Three filled and sealed tray pack cans shall be selected at random from the lot. The tray pack cans shall be individually tested for fat and salt content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, chapter: Meat and Meat Products, except that preparation of the samples shall be as follows: The unopened tray pack cans shall be gently warmed in a water bath to melt fat adhering to the inside of the cans. The cans shall be opened and the entire contents of each can shall be separately blended in a Waring Blendor or equivalent. The test results shall be reported to the nearest 0.1 percent. Any result failing to conform to the fat and salt requirements in 3.6 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.4 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers, except that inspection for labeling shall be as specified in 4.5.4.1. In addition, the following defect shall be classified as a major defect:

Evidence of buffing with an abrasive substance (see 5.1.1)

4.5.4.1 Can label examination. Labels shall be examined for defects in accordance with MIL-L-1497 (see 5.4) except, for self-adhering labels, the following additional defects shall apply:

Major: Label torn or scratched so as to obliterate any of the markings.

Minor: Air bubbles under label.
Label not properly adhered to can, for example, label raised or peeled back from edges or corners.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:28Z
Check the source to verify that this is the current version before use.

4.5.4.2 Label adhesive examination. When self-adhering labels are used, the adhesive shall be tested in accordance with ASTM D 3330.

4.5.5 Can closure examination. Can closure shall be examined visually and by teardowns in accordance with the can manufacturer's requirement and 21 CFR, Part 113, Subpart D, or 9 CFR, Part 318, Subpart G, as applicable. Any nonconformance based on observation of can seam teardowns or on record of can seam teardowns shall be classified as a major defect and shall be cause for rejection of any involved product.

4.5.6 Vacuum examination. Cans shall be allowed to cool to 75° ± 5°F, held for at least 24 hours after sealing, and then examined for vacuum retention. To examine, lay a straight edge in the center of the lid along the length of the tray pack. Both ends of the straight edge shall touch the lid at the inside edge of the double seam. There shall be a visible gap between the straight edge and the lid for the entire distance of the label panel. Using a shorter straight edge, the same procedure shall be used across the width, in the center of the tray pack can. One measurement shall be made when examining a ribbed lid; lay the straight edge between the two center ribs along the length of the can. The inspection lot shall include only tray packs produced in a single shift on a single sealing machine. The sample size shall be 50 cans. Any nonconformance shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.7 Test for rust spots on interior of tray pack cans. Where rust spots are suspected on interior of tray pack cans, the following test shall be performed:

a. Immerse a cotton swab in acetone or methyl ethyl ketone solvent and gently rub suspected spot. Handling of the solvent shall be in strict accordance with the guidelines of the manufacturer's Material Safety Data Sheet (MSDS).

b. If suspected spot immediately disappears, it is not a rust spot.

c. If suspected spot remains, continue gently rubbing with the swab resoaked with solvent, and observe for disappearances or retention of spot. Replenish solvent as necessary.

d. If spot persists, and all but a very thin coating, or no coating remains, it will be scored as a rust spot. This can be confirmed further by rubbing with a finger. The rust spot will be felt as a slight bump over the substrate.

4.5.8 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:28Z
Check the source to verify that this is the current version before use.

Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible.
Reinforced with other than nonmetallic strapping or tape.
Dimensions of pads not as specified.
Interior packing with fiberboard liner or pads not as specified.

Minor: Other required markings missing, incorrect, or illegible.
Arrangement or number of cans not as specified.

4.5.9 Unit load inspection.. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PACKAGING

5.1 Preservation. The product shall be preserved in accordance with level A.

5.1.1 Level A. One hundred and six ounces of food product shall be filled into a tray pack can conforming to MIL-C-44340 and sealed and thermoprocessed as specified in 3.4 and 3.5. The practice of reconditioning tray pack cans by buffing with an abrasive substance shall not be permitted (see 4.5.4). The interior coating of filled and sealed thermoprocesed cans shall be free of rust spots, scratches, or fractures that penetrate through the coating when examined in accordance with 4.5.3.2.

5.2 Packing. The product shall be packed in accordance with level A, B, or C, as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L or HSC-L with an HSC full depth cover, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads fabricated of grade V3c fiberboard. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box. Each box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament-reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1), except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern so as to create columnar stacking. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:28Z
Check the source to verify that this is the current version before use.

5.2.2 Level B packing. Four cans of product, preserved as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.3 Level C packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of the same material as the box.

5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 or 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified, except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern so as to create columnar stacking. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497 and with the following:

- Official establishment number (for example, EST 38) or a three-letter code identifying the establishment
- Lot number 1/
- Production shift number 1/
- Retort identification number 1/
- Retort cook number 1/

1/ The lot number shall be expressed as a four-digit Julian code. The first digit shall indicate the year of production and the next three digits shall indicate the day of the year (Example, March 19, 1990, would be coded as 0078). The Julian code shall represent the day the product was packaged and processed. Sub-lotting (when used) shall be represented by an alpha character immediately following the four-digit Julian code. Following the four-digit Julian code and the alpha character (when used), the other required code information shall be printed in the sequence as listed above.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:28Z
Check the source to verify that this is the current version before use.

In addition, the name of the product shall be marked, stamping is permitted, on one 1001 by 200 side of the can. The labeling shall be legible when examined as specified in 4.5.4 after preparation of product in accordance with heating instructions. Paper labels are not permitted. Cans shall show the following statements:

TO HEAT IN WATER: Submerge unopened can in boiling water. Simmer gently 40 to 45 minutes. Avoid overheating (can shows evidence of bulging).

CAUTION: Use care when opening as pressure may have been generated within the can.

TO HEAT IN OVEN: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place. Place in a 350°F oven 35 to 40 minutes.

WARNING: Do not place unopened can in oven. This may cause the can to burst.

YIELD: Serves 18 portions of 2/3 cup each.

As an alternative labeling method, a preprinted, self-adhering, 0.002-inch thick, clear polyester label printed with indelible black ink may be used. Self-adhering labels shall be applied after retorting. Pressure-sensitive adhesive shall require no preparation prior to application. Labels shall tack quickly and adhere without curling or breaking. The adhesive shall have a minimum adhesion of 60 ounces per inch width when examined as specified in 4.5.4.2. When self-adhering labels are used, the tray pack cans shall be labeled with the Julian code and a product code prior to retorting.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary marking in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT OR MORE THAN TWO HIGH IN STORAGE, UNLESS PALLET RACKS ARE USED.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:28Z
Check the source to verify that this is the current version before use.

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.4, and 6.2).
d. Provisions for approved preproduction samples (see 3.6.1 and 6.2).
e. Level of packing required (see 5.2).
f. Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.4 Subject term (key word) listing.

Canned foods
Combat field feeding
Operational rations
Poultry
Shelf stable

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

(Project 8940-0698)

Review activities:

Army - MD, QM
Navy - MC
DP - SS

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:28Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-44328A	1990 July 11

3. DOCUMENT TITLE

CHICKEN CACCIATORE, THERMOSTABILIZED, TRAY PACK

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY		
a. NAME U.S. Army Natick RD&E Center	b. TELEPHONE (Include Area Code) (1) Commercial 508-651-5221	(2) AUTOVON 256-5221
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-ES Natick, MA 01760-5014	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340	AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-22 16:28Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T01:28Z
Check the source to verify that this is the current version before use.